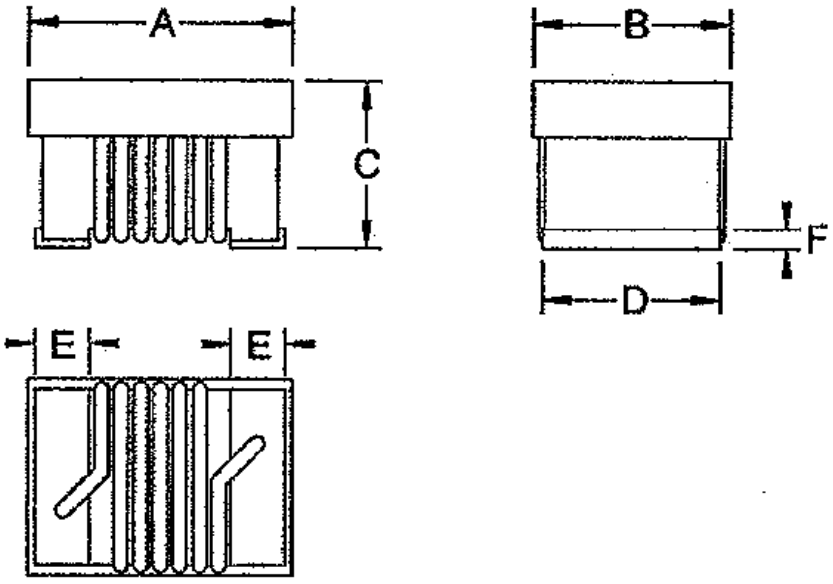
INCH-POUND
MIL-PRF-83446/36B
w/AMENDMENT 3
2 September 2025
SUPERSEDING
MIL-PRF-83446/36B
w/AMENDMENT 2
4 May 2020

PERFORMANCE SPECIFICATION SHEET

COILS, RADIO FREQUENCY, CHIP, OPEN CONSTRUCTION, SURFACE MOUNT, 0603 SIZE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-83446.



	INCHES	MILLIMETERS
A	0.071 Max.	1.80 Max.
B	0.047 Max.	1.19 Max.
C	0.040 Max.	1.02 Max. For Termination finish A
	0.043 Max.	1.09 Max. For Termination finish B or F
D	0.030 (Ref. Only)	0.76 (Ref. Only)
E	0.018 (Ref. Only)	0.44 (Ref. Only)
F	0.008 (Ref. Only)	0.20 (Ref. Only)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.

AMSC N/A
FSC 5950
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TABLE I. Electrical characteristics (initial).

Dash Number	Inductance nH 1/	Inductance Tolerance Percent (±)	Inductance Test Frequency MHz	Q (min) 1/	Q Test Frequency (MHz)	Self Resonant Frequency MHz (min) 2/	DC Resistance Ohms (max)	Current Rating mA (max)
-01	1.8	2, 5, 10	250	22	250	6000	0.07	1000
-02	3.6	2, 5, 10	250	22	250	6000	0.07	1000
-03	3.9	2, 5, 10	250	22	250	6000	0.07	1000
-04	4.7	2, 5, 10	250	22	250	6000	0.09	885
-05	6.8	2, 5, 10	250	30	250	5800	0.10	840
-06	10	2, 5, 10	250	31	250	4800	0.16	665
-07	12	2, 5, 10	250	35	250	4000	0.13	735
-08	15	2, 5, 10	250	35	250	4000	0.17	645
-09	18	2, 5, 10	250	35	250	3100	0.17	645
-10	22	2, 5, 10	250	38	250	3000	0.19	610
-11	27	2, 5, 10	250	40	250	2800	0.22	565
-12	33	2, 5, 10	250	40	250	2300	0.22	565
-13	39	2, 5, 10	250	40	250	2200	0.24	540
-14	47	2, 5, 10	200	38	200	2000	0.28	500
-15	56	2, 5, 10	200	38	200	1900	0.31	475
-16	68	2, 5, 10	200	37	200	1700	0.36	440
-17	82	2, 5, 10	150	34	150	1700	0.54	360
-18	100	2, 5, 10	150	34	150	1400	0.75	305
-19	120	2, 5, 10	150	32	150	1300	0.79	300
-20	150	2, 5, 10	150	28	150	990	1.14	245
-21	180	2, 5, 10	150	25	150	990	1.28	235
-22	220	2, 5, 10	100	25	150	700	1.70	200
-23	270	2, 5, 10	25	16	25	600	1.78	195
-24	3.3	2, 5, 10	250	22	250	6000	0.07	1000
-25	4.3	2, 5, 10	250	22	250	6000	0.09	885
-26	5.6	2, 5, 10	250	22	250	5800	0.10	840
-27	8.7	2, 5, 10	250	30	250	5200	0.14	705
-28	16	2, 5, 10	250	35	250	3400	0.17	645

1/ Inductance and Q are tested using HP4291A or HP4287A with test fixture 16193A, or equivalent.
2/ Self resonant Frequency tested using HP8753C, or equivalent.

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TABLE II. Electrical characteristics (final).

Inspection group	Allowable percent variation			
Qualification Inspection:	Inductance 1/	Q 2/	Self-resonant Frequency 2/	DC resistance 1/
Group II	±5	-10	-8	±3 (+.001 ohm)
Group IV	±5	-10	-10	±3 (+.001 ohm)
Group V	±5	-10	-----	-----
Quality Conformance Inspection:				
Group C				
Subgroup II	±5	-10	-10	±2 (+.001 ohm)
Subgroup IV	±5	-10	-8	±3 (+.001 ohm)
Subgroup V	±5	-10	-----	-----

1/ From initial measurement.
2/ From specified minimum value in electrical characteristics (initial) table.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Bond strength: Force 1 pound.

Weight: 0.008-gram, maximum.

Operating temperature range: -55°C to +125°C.

Temperature rise (at +90°C): +35°C.

Maximum operating temperature: +125°C.

Altitude: 70,000 feet.

Solderability: Test method S of J-STD-002, except solder preforms may be used or MIL-STD-202-208, test method B. Both terminals are immersed simultaneously. Magnet wire and bond are excluded from solder coverage.

Dielectric withstanding voltage: MIL-STD-202-301, test voltage 200 Vrms for 5 seconds for quality conformance.

Barometric pressure: MIL-STD-202-105, test condition C, (70,000 feet), test voltage 80 Vrms.

DWV, Barometric pressure and Insulation resistance tests are to be performed with units mounted to a test substrate. Tests are after visual/mechanical inspection of qualification Group I, and after electrical characteristics (initial) of subgroup II Group C.

Electrical characteristics (initial): See table I.

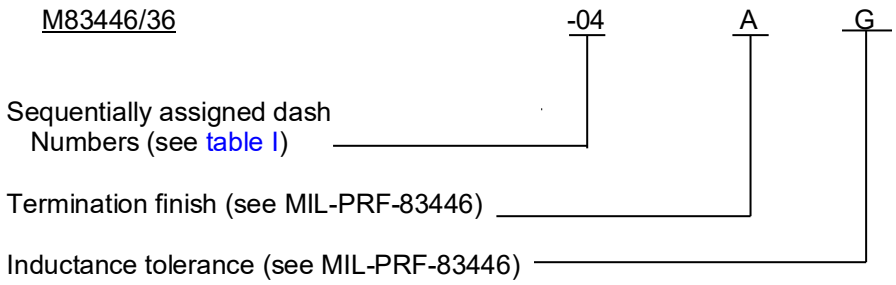
Electrical characteristics (final): See table II.

Vibration: MIL-STD-202-204, test condition D. During testing, each part to be monitored for electrical continuity. Any discontinuity exceeding 100 microseconds shall be cause for rejection.

Mechanical shock: MIL-STD-202-213, test condition I. After the test, parts are to be inspected for mechanical damage and verified for electrical continuity.

Vibration and Mechanical shock testing shall be performed as part of Group V tests for qualification (before moisture resistance) and as part of subgroup V (semiannually) for group C testing. Parts shall be mounted on test substrates.

Part or Identifying Number (PIN): The PIN shall be in the following form:



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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-83446, this document references the following:

MIL-STD-202-105
MIL-STD-202-204
MIL-STD-202-208
MIL-STD-202-213
MIL-STD-202-301
J-STD-002

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - MI
Navy - AS, OS
Air Force - 19

Preparing activity:
DLA - CC

(Project 5950-2025-043)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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